

Predicting Remaining Useful Life in Commercial Turbofans: A Deep-Learning Pipeline for Variable-Length N-CMAPSS Flights

Prahalad Talur

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Abstract

Predicting the remaining useful life (RUL) of commercial turbofan engines from in-flight sensor recordings is a core problem in aerospace prognostics: an accurate, well-calibrated estimator enables condition-based maintenance, reduces unscheduled removals, and bounds the operational risk of in-service failure. The N-CMAPSS dataset [1] replaces the synthetic CMAPSS benchmark [14] with run-to-failure trajectories generated under real recorded flight envelopes, yielding variable-length flights of 10^4 – 10^5 timesteps per engine and heterogeneous operating conditions that previously published deep models were not designed to ingest. We present a deep-learning pipeline for variable-length N-CMAPSS flights and use it to compare recurrent, temporal-convolutional, transformer, and multi-scale attention families under the same preprocessing, training, and evaluation code. Operationally short windows of roughly one thousand timesteps are competitive with, and in our benchmarks substantially better than, full-flight inputs; the gap is confounded with architecture, but the pattern is consistent with the short-window choices of prior PHM-2021 challenge submissions [15, 12]. A multi-scale TCN with global-fusion attention (MSTCN) sits in the top performance cluster alongside transformer and WaveNet baselines, is best by RMSE on FD2, and outperforms full-flight RNN baselines whose R^2 is negative. We use asymmetric MSE with a $2\times$ late-prediction penalty as a design choice anchored on the asymmetric PHM scoring function [14] and recent loss-design work [11], rather than as an empirical contribution. On the apples-to-apples FD1 benchmark, the best controlled model attains RMSE 6.523 cycles, within the band reported by N-CMAPSS prior art.

1 Introduction

Modern commercial turbofans are instrumented with dense sensor suites whose readings, integrated over thousands of flight hours, encode the gradual degradation of compressor, turbine, and combustor components. Translating those recordings into a calibrated estimate of remaining useful life is the central task of data-driven engine prognostics: a reliable RUL estimate lets airlines plan maintenance against actual wear rather than fixed intervals, lowers the rate of unscheduled removals, and bounds the operational risk of late detection of an evolving fault. The cost structure of this task is sharply asymmetric. Predicting RUL too early is conservative and incurs maintenance overhead; predicting it too late risks an in-service failure. The PHM-2008 scoring function [14] encodes this asymmetry directly and remains the canonical evaluation metric in the field.

For more than a decade, deep-learning research on this problem has been driven by the synthetic CMAPSS benchmark [14], where each engine produces a few hundred cycles of low-dimensional sensor data under a small set of operating conditions. The N-CMAPSS dataset of Arias-Chao et al. [1] was introduced to close the gap between this benchmark and the conditions encountered in commercial operation: trajectories are simulated from real recorded flight envelopes spanning

ascent, cruise, and descent, with sample-rate-level sensor data on the order of 10^4 – 10^5 timesteps per cycle. The hybrid physics-augmented baseline of Arias-Chao et al. [2] and the published submissions to the 2021 PHM Society Data Challenge [12, 6, 15] establish N-CMAPSS as the contemporary benchmark of record for turbofan RUL.

N-CMAPSS poses three difficulties that did not arise on CMAPSS. Flights are variable-length, with each engine yielding a different total cycle count and per-cycle sequence length. Operating conditions are heterogeneous within and across flights, so a single model must generalize across the full envelope. And the sequence regime is qualitatively different: at 10^4 – 10^5 timesteps per cycle, naively ingesting full flights pushes recurrent and quadratic-attention models toward their failure modes for long inputs.

The methodological response to these difficulties has spanned four families. Recurrent networks such as LSTM [8], GRU [4], and bidirectional variants were the first applied. Temporal convolutional networks [3], WaveNet-style dilated convolutions [16], and dilated/inception variants [12, 6] were introduced to handle long inputs without sequential bottlenecks. Self-attention models, building on the Transformer [17] and long-sequence variants such as Informer [19], have been adapted to RUL through hierarchical and two-stage designs [7, 10]. Most recently, multi-scale TCN architectures with attention modules over the dilation hierarchy [18] have been proposed as a way to combine the long effective receptive field of TCNs with the cross-channel and cross-scale weighting of attention. The publicly reported benchmarks across these families are not directly comparable: most are reported on CMAPSS rather than N-CMAPSS, training protocols and input lengths vary substantially, and few studies report the same accuracy, RMSE, and PHM-score triple under a single harness.

This paper contributes a deep-learning pipeline for variable-length N-CMAPSS flights together with a shared registry of 22 architectures drawn from the main recurrent, convolutional, hybrid, transformer, and multi-scale attention families. The pipeline standardizes sequence preparation (last- N truncation, per-feature standardization), the loss function (asymmetric MSE with a $2\times$ late-prediction penalty), the evaluation metrics (RMSE, R^2 , PHM score, accuracy at ± 20 cycles), and the apples-to-apples experimental harness used for the head-to-head numbers. The results support the following claims:

- **C1 (sequence length).** Operationally short windows of roughly one thousand timesteps are competitive with, and in our benchmarks substantially better than, full-flight inputs. Full-flight RNN baselines yield negative R^2 (T8: LSTM RMSE 22.28, $R^2 = -0.066$; T20: LSTM RMSE 22.28; T21: BiLSTM RMSE 22.20) while short-window models cluster at RMSE 6.5–7.6 (T1–T3); the magnitude is confounded with architecture choice and we report it as a benchmark ceiling rather than as a controlled single-model sweep.
- **C2 (multi-scale temporal attention).** A multi-scale TCN with global-fusion attention (MSTCN) sits in the top cluster of attention and multi-scale models on N-CMAPSS, is best on FD2 by RMSE (T11: 6.495), and is far ahead of the full-flight RNN baselines. We do not claim MSTCN beats every alternative: in the apples-to-apples FD1 benchmark MSTCN ranks third (T3: 7.604) behind WaveNet (T1: 6.523) and CNN-GRU (T2: 7.006).
- **C3 (asymmetric loss).** We adopt asymmetric MSE with a $2\times$ late-prediction penalty as a methodological choice anchored on the PHM scoring asymmetry of Saxena et al. [14] and the recent multiple-penalty-mechanism loss of [11], not as an empirical contribution. A controlled symmetric-MSE baseline and an α -sensitivity sweep remain to be run.

The rest of the paper is organized as follows. Section 2 reviews related work in N-CMAPSS prognostics, attention-based RUL models, and asymmetric loss design. Section 3 describes the

pipeline, including data handling, the asymmetric loss, and the MSTCN architecture. Section 4 presents the apples-to-apples benchmark across the registered architecture set. Section 5 discusses C1–C3 and the missing ablations. Section 6 concludes.

2 Related Work

We organize prior work into four families directly relevant to the claims in this paper: the N-CMAPSS dataset and its baseline competitions, attention/transformer architectures for RUL, multi-scale temporal-convolution methods, and asymmetric losses for safety-critical prognostics. We close by situating our contribution against this literature, noting up front that nearly all of the attention- and multi-scale architectures we benchmark have, to date, been published only on the older C-MAPSS dataset and that head-to-head comparability with our N-CMAPSS results is therefore limited.

2.1 N-CMAPSS dataset and baseline competitions

The N-CMAPSS dataset [1] extends the original C-MAPSS turbofan simulator [14] to run-to-failure trajectories generated under recorded flight conditions, producing variable-length flights on the order of 10^4 – 10^5 timesteps per unit. The 2021 PHM Society Data Challenge introduced the first comparable leaderboard on this dataset. The challenge winner [12] proposed a dilated-convolutional network explicitly designed to ingest variable-length input sequences with a learned condition-normalization step, which is the closest prior art to the variable-length flight handling we adopt. The second-place entry [6] used inception modules to capture multiple temporal receptive fields from windowed N-CMAPSS data, and the third-place entry [15] reported a stacked DCNN trained on $L_w = 161$ windows, achieving validation L2 RMSE 6.24 and test NASA score 3.651. Outside the challenge, a hybrid physics-plus-deep-learning baseline [2] reports an extended prediction horizon over a purely data-driven model on a fleet of nine N-CMAPSS engines, and the ANN-Flux baseline of Cohen et al. [5] reports RMSE 7.75 / NASA score 4.34 on N-CMAPSS. These are the primary numerical anchors for “what is a good N-CMAPSS result,” and they collectively establish a precedent that compact windowed inputs are the operating regime of choice on this dataset.

2.2 Attention and transformer architectures for RUL

The transformer [17] replaces recurrence with self-attention as the primary sequence-modeling primitive, and Informer [19] adapts it to long sequences via ProbSparse attention and self-attention distillation, motivating input compression rather than raw long inputs. Two recent attention models are directly relevant. The STAR transformer [7] is a two-stage hierarchical encoder–decoder that applies sequential sensor-wise and temporal attention; it operates on 32–64-step inputs and reports RMSE 10.61 / 13.47 / 10.71 / 15.87 on C-MAPSS FD001–FD004. TTSNet [10] fuses a transformer branch, a TCN branch and a multi-head self-attention branch over a noise-smoothed input and reports RMSE 11.02 / 13.25 / 11.06 / 18.26 on the same C-MAPSS subsets. Both are evaluated only on C-MAPSS, so their RMSE numbers are not directly comparable to ours: N-CMAPSS sequences are one to two orders of magnitude longer per unit, and the resulting RUL distributions differ.

2.3 TCN and multi-scale temporal methods

The temporal convolutional network [3] establishes dilated causal convolutions as a competitive sequence-modeling alternative to recurrent networks, with longer effective memory and faster train-

ing. The closest prior work to our headline architecture is the attention-based multi-scale TCN of Xu, Zhang and Miao [18], which combines a multi-scale TCN backbone with self-attention at the head and a global-fusion-attention block at the tail. Their architecture is essentially the same family our paper places in the “MSTCN” slot, and is therefore the strongest direct precedent for our multi-scale temporal-attention claim. As with the transformer literature, however, [18] reports results on C-MAPSS rather than N-CMAPSS, so the architectural lineage is direct but the numerical comparison with our results is not.

2.4 Loss functions for safety-critical RUL

The dominant evaluation metric for RUL on C-MAPSS and N-CMAPSS is the asymmetric exponential PHM score introduced by Saxena et al. [14], which penalizes late predictions exponentially harder than early ones (canonical penalty constants 13 and 10). Training-time loss functions that mirror this asymmetry have been proposed in recent work; the multiple-penalty-mechanism loss of [11] pairs RMSE with similarity-based late-prediction penalties and reports improved advanced-prediction probability on C-MAPSS subsets. We adopt asymmetric MSE with a $2\times$ late-prediction penalty as a methodological choice anchored on [14] and consistent with [11], rather than as an empirical contribution.

2.5 Where this work fits

Two observations motivate our contribution. First, recent attention and multi-scale-TCN architectures for RUL [7, 10, 18] are reported almost exclusively on the older C-MAPSS dataset, even though N-CMAPSS [1] has been available since 2021 and exposes substantially longer, multi-condition, variable-length flights. Second, the published N-CMAPSS literature itself [12, 6, 15, 5, 2] consists of a small number of competition entries and hybrid baselines, with no broad architecture sweep under matched preprocessing and training. Our work therefore benchmarks attention, multi-scale-TCN, transformer, WaveNet and RNN baselines on N-CMAPSS itself; the limited apples-to-apples N-CMAPSS attention literature is itself part of what makes the sweep informative. Where we cite C-MAPSS-only RMSE numbers (e.g. [7, 10, 18]), we do so as architectural precedent, not as a direct numerical baseline.

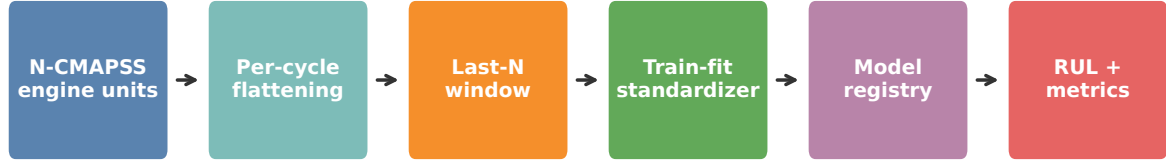
3 Methods

3.1 Dataset and task

We evaluate remaining-useful-life regression on N-CMAPSS [1], using the train, validation, and test splits exposed by the `rul_datasets` reader [13]. Each engine unit is represented as a list of per-cycle multivariate sensor sequences, and each cycle has one scalar RUL target. The pipeline treats each cycle as one supervised example with input shape $(T, 32)$, where T is the retained temporal length and 32 is the number of sensor channels. FD1 is the primary controlled benchmark and FD2 is used as a cross-subset robustness check.

3.2 Variable-length sequence preparation

Raw N-CMAPSS cycles can contain long, variable-length flight records. The training pipeline flattens the per-engine lists into per-cycle examples and, when a maximum sequence length is specified, keeps the last T timesteps of each cycle. Unless otherwise noted, the apples-to-apples benchmark



Common preprocessing, loss, callbacks, and evaluation harness across model families

Figure 1: Preprocessing and evaluation pipeline. Each engine is flattened into per-cycle examples, truncated to the last N timesteps, normalized with train-fit feature statistics, passed through the shared model registry, and evaluated with the same RUL metrics.

uses $T = 1000$. This last-window convention encodes the hypothesis that recent degradation behavior is more predictive of near-term RUL than distant early-flight history. After truncation, feature normalization is fit on the training split only and applied unchanged to validation and test examples, avoiding leakage from held-out engines.

3.3 Loss and metrics

All controlled benchmark rows in the main comparison use asymmetric MSE with late-prediction multiplier $\alpha = 2.0$:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \begin{cases} \alpha(\hat{y}_i - y_i)^2, & \hat{y}_i - y_i \geq 0, \\ (\hat{y}_i - y_i)^2, & \hat{y}_i - y_i < 0. \end{cases} \quad (1)$$

Here over-predicting RUL is treated as a late maintenance decision and is penalized more strongly than conservative under-prediction. This is a design choice motivated by the asymmetric PHM scoring function of Saxena et al. [14] and recent asymmetric-loss work [11]; Section 5 makes clear that we do not yet have a controlled symmetric-MSE baseline.

Evaluation reports RMSE, R^2 , PHM score, and threshold accuracies. Accuracy@20 is the percentage of predictions within ± 20 cycles of the true RUL. The PHM score follows the standard asymmetric exponential form, with late errors penalized more sharply than early errors.

3.4 Model registry and benchmark harness

The repository currently registers 22 neural architectures under a common registry interface. The catalog spans a no-temporal MLP baseline, recurrent models [8, 4], dilated convolutional models [3, 16], CNN-RNN hybrids, attention models [17], transformer variants [7], and recent multi-scale TCN/fusion architectures [10, 18]. This shared interface lets the same preprocessing, optimizer, callbacks, loss, and metric code drive every architecture in the benchmark. Standard hyperparameters are units=64, dense units=32, dropout=0.2, Adam [9] with learning rate 0.001, 30 epochs for the controlled runs, and early stopping on validation loss.

3.5 MSTCN architecture

The flagship model is a multi-scale temporal convolutional network with global fusion attention (MSTCN), following the architectural family of attention-based multi-scale TCNs [18]. Given an

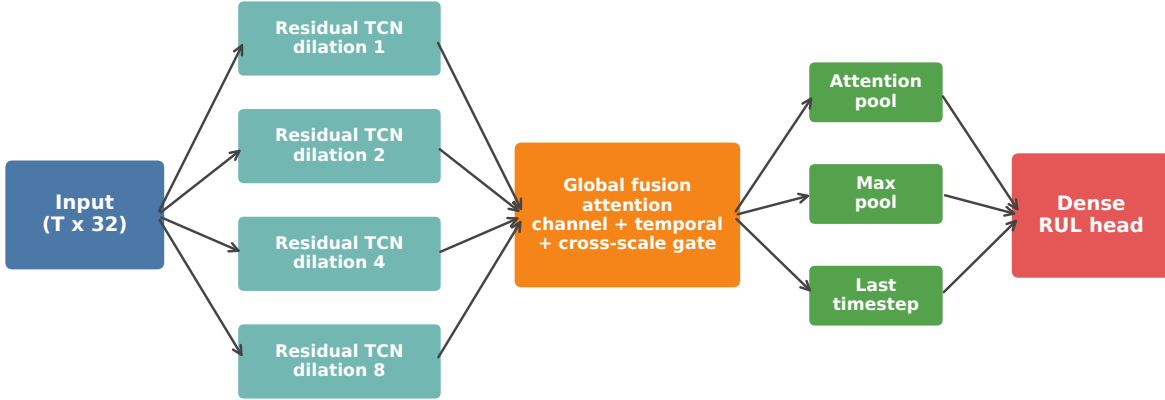


Figure 2: MSTCN architecture used in the benchmark. Parallel residual TCN branches cover four dilation scales before global fusion attention combines channel, temporal, and cross-scale information for the final RUL regression head.

input sequence $(T, 32)$, MSTCN applies four parallel residual TCN branches with dilation rates 1, 2, 4, and 8. Each branch captures a different temporal receptive field over the same sensor stream. The branch outputs are then passed through a global fusion attention block that combines channel attention, temporal attention, cross-scale concatenation, and a learned fusion gate. The fused representation is pooled over time and passed through a dense regression head to produce one scalar RUL estimate.

For the default configuration, MSTCN uses 64 filters per branch and a 32-unit dense head, yielding roughly 150k parameters. This is larger than some baselines, so our fixed-protocol claim is limited to “competitive” rather than “superior at matched parameter count.” An iso-parameter comparison has not yet been run.

4 Experiments

4.1 Experimental setup

The controlled benchmark uses N-CMAPSS FD1 as the primary subset and FD2 as a cross-subset check. Unless otherwise noted, the apples-to-apples rows use the same data pipeline described in Section 3: last-1000 timestep truncation, train-fit feature standardization, asymmetric MSE with $\alpha = 2.0$, Adam optimization [9], and the shared callback stack. We assign stable row identifiers (T1, T2, ...) to the canonical results so that the discussion can refer to exact experimental records without repeating the full configuration each time. Tables and figures that report our own results cite these row identifiers for provenance; external comparison rows are cited to their source papers.

4.2 Architecture sweep

Table 1 summarizes the architecture families covered by the registry. The current codebase catalog contains 22 registered models, ranging from simple no-temporal baselines to multi-scale attention and transformer variants.

Table 1: Architecture families in the benchmark harness.

Family	Registered models	Role in this paper
Baseline	MLP	No-temporal lower bound
RNN	LSTM [8], BiLSTM, GRU [4], BiGRU	Long-sequence recurrent baselines
Convolutional	TCN [3], WaveNet [16]	Dilated convolution baselines
Hybrid	CNN-LSTM, CNN-GRU, Inception-LSTM	Local feature extractor plus recurrence
Attention	Attention-LSTM, Transformer [17], ResNet-LSTM	Attention and transformer comparisons
SOTA/fusion	MDFA, MDFA-paper, CNN-LSTM-Attention, CATA-TCN, TTSNet [10], ATCN, MSTCN [18]	Multi-scale / attention frontier
Transformer variants	STAR [7], sparse-transformer-BiGRU	Recent hierarchical/sparse attention

Table 2: Canonical FD1 results. T1–T3 use the apples-to-apples harness ($T=1000$, 30 epochs). T8 and T10 are lower-bound baselines with different protocols (full-flight and no-temporal, respectively).

Row	Model	RMSE	R^2	PHM	Acc@20
T1	WaveNet	6.523	0.9086	0.7256	98.83
T2	CNN-GRU	7.006	0.8946	0.8396	98.53
T3	MSTCN	7.604	0.8758	1.0033	98.53
T8	LSTM, full flight ($T=20294$)	22.28	-0.0657	–	–
T10	MLP ($T=1000$)	17.45	0.3461	–	–

4.3 FD1 headline results

Table 2 reports the controlled FD1 head-to-head. WaveNet is the strongest apples-to-apples FD1 model by RMSE (T1: 6.523), followed by CNN-GRU (T2: 7.006) and MSTCN (T3: 7.604). MSTCN is therefore not the FD1 winner under the most rigorous controlled protocol.

The FD1 result is a cluster result: dilated convolution, hybrid, and multi-scale attention models operate in the 6.5–7.6 RMSE band (T1–T3), while no-temporal and full-flight recurrent baselines are far worse (T8, T10). This supports the claim that attention/multi-scale temporal models are competitive on N-CMAPSS, while MSTCN’s FD1 rank remains below WaveNet and CNN-GRU.

4.4 FD2 robustness

On FD2, MSTCN is the best model by RMSE among the available apples-to-apples rows (T11: 6.495), with WaveNet close behind (T12: 6.739) and CNN-GRU degrading under the more variable flight envelope (T13: 19.812). WaveNet still has the highest Accuracy@20 on FD2 (T12: 99.50 versus T11: 99.01). Thus, the FD2 result supports the “top cluster” framing for MSTCN without establishing dominance across every metric.

4.5 Multiseed reproducibility

To bound run-to-run variance we trained MSTCN over three random seeds (42–44) and WaveNet over two seeds (45–46) on FD1 with the same apples-to-apples protocol. Table 4 reports the model-level means. The MSTCN mean RMSE of 8.140 (± 0.74 , $n=3$) and WaveNet mean of 8.238 (± 1.44 , $n=2$) are both above the best-run figures in Table 2 (T1: 6.523; T3: 7.604), confirming that single-seed results reflect favorable draws. WaveNet’s wider standard deviation (± 1.44) indicates higher run-to-run variability than MSTCN (± 0.74); with only two to three seeds, the two models’ multiseed means are not meaningfully separable.

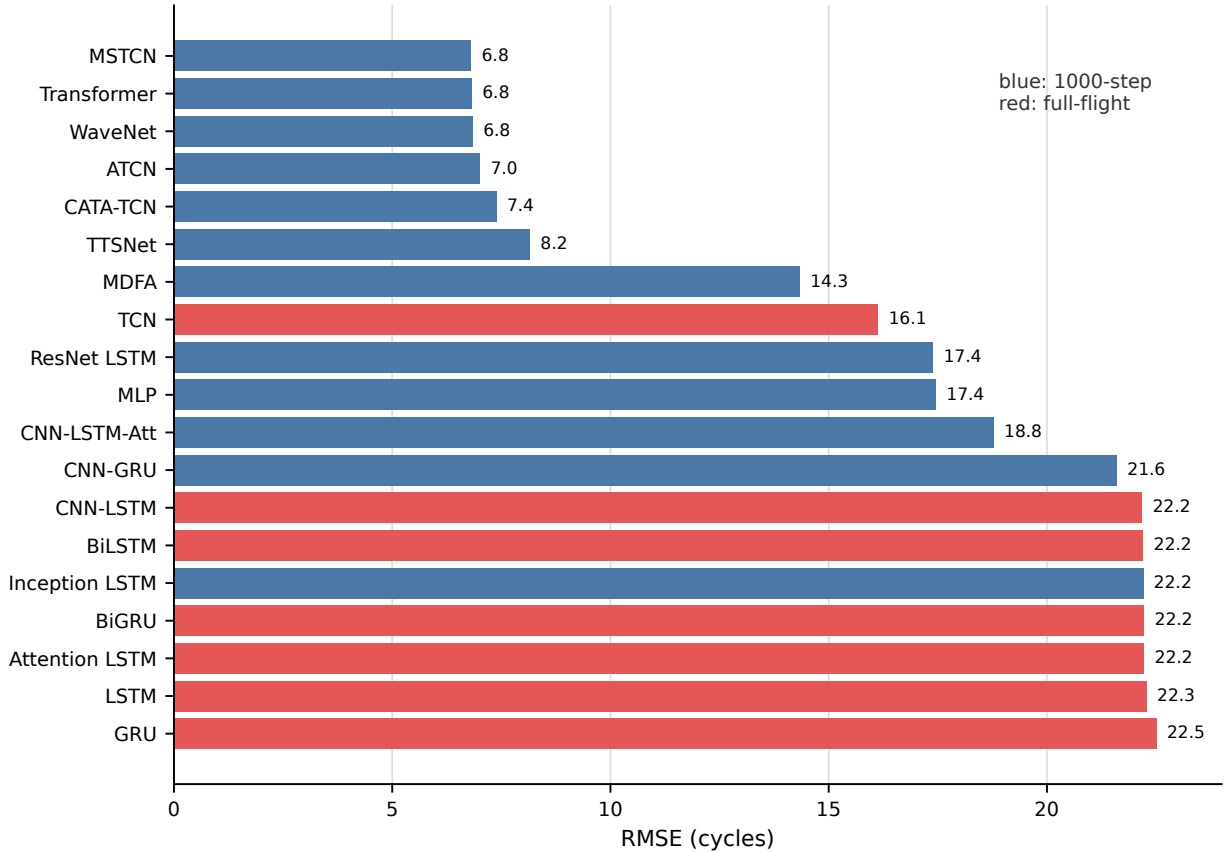


Figure 3: Historical FD1 architecture sweep from the repository comparison artifact. The figure is used as broad architecture-family context for recurrent [8, 4], convolutional [3, 16], transformer [17, 7], and multi-scale attention models [10, 18]; the controlled head-to-head claims in the paper use the apples-to-apples rows in Tables 2–3.

4.6 Sequence-length evidence

Table 5 gives the available evidence behind the sequence-length discussion. The comparison is deliberately labelled as evidence rather than a complete ablation: the 1000-timestep rows are controlled head-to-head runs, the 250-timestep rows are shorter 10-epoch probes, and the full-flight rows come from older full-length baselines. The table still shows a clear practical pattern: the best 1000-timestep models are substantially stronger than the 250-timestep and full-flight rows, but the architecture and training differences prevent a causal sequence-length claim.

4.7 External comparison

The closest N-CMAPSS literature anchors are Solís-Martín et al. [15], who report validation RMSE 6.24 using a 161-cycle window, and Cohen et al. [5], who report RMSE 7.75 for ANN-Flux. Our controlled WaveNet FD1 result (T1: 6.523), controlled MSTCN FD1 result (T3: 7.604), and controlled MSTCN FD2 result (T11: 6.495) fall within this N-CMAPSS band. C-MAPSS-only results from STAR [7], TTSNet [10], and Xu et al. [18] are treated as architectural precedent rather than direct numerical baselines because the datasets and sequence regimes differ substantially.

We dissect these results across sequence length, architecture family, and loss form in Section 5.

Table 3: Canonical FD2 cross-subset results.

Row	Model	RMSE	R^2	PHM	Acc@20
T11	MSTCN	6.495	0.8897	0.5755	99.01
T12	WaveNet	6.739	0.8813	0.6756	99.50
T13	CNN-GRU	19.812	-0.0265	5.1463	59.41

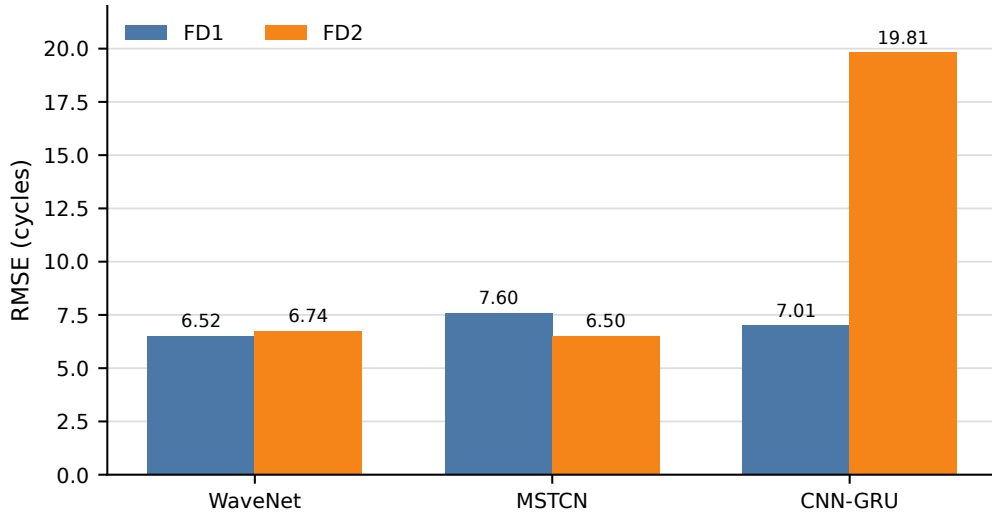


Figure 4: Controlled RMSE comparison on FD1 and FD2 for the three models available in both apples-to-apples runs. WaveNet is best on FD1, MSTCN is best by RMSE on FD2, and CNN-GRU degrades sharply on FD2.

5 Ablations and Discussion

This section interprets the benchmark against the three paper claims. Several experiments needed for causal attribution are absent, so we separate supported observations from missing ablations.

5.1 C1: sequence length

The benchmark supports the directional claim that short operational windows are effective on N-CMAPSS. At 1000 timesteps, WaveNet reaches RMSE 6.523 on FD1 (T1), MSTCN reaches RMSE 7.604 in the same FD1 protocol (T3), and MSTCN reaches RMSE 6.495 on FD2 (T11). Full-flight baselines are much weaker: the full-flight TCN row reports RMSE 16.13 (T19), while full-flight LSTM and BiLSTM rows report RMSE 22.28 (T20) and 22.20 (T21). Very short 250-step inputs also degrade performance, with MSTCN at RMSE 10.748 (T14) and WaveNet at RMSE 11.169 (T15), suggesting that the useful regime is not simply “as short as possible.”

The limitation is that no single architecture has been trained across the full length sweep $\{100, 250, 500, 1000, 2000, 5000, 20000\}$ under matched conditions. The 1000-step winners are mostly attention or dilated-convolution models, whereas the 20k rows are mostly recurrent or simpler convolutional baselines. The apparent 1k-vs-full gap therefore conflates sequence length with architecture family. We report C1 as a directional finding, corroborated by short-window N-CMAPSS challenge submissions [12, 15], not as a controlled 58% single-factor ablation.

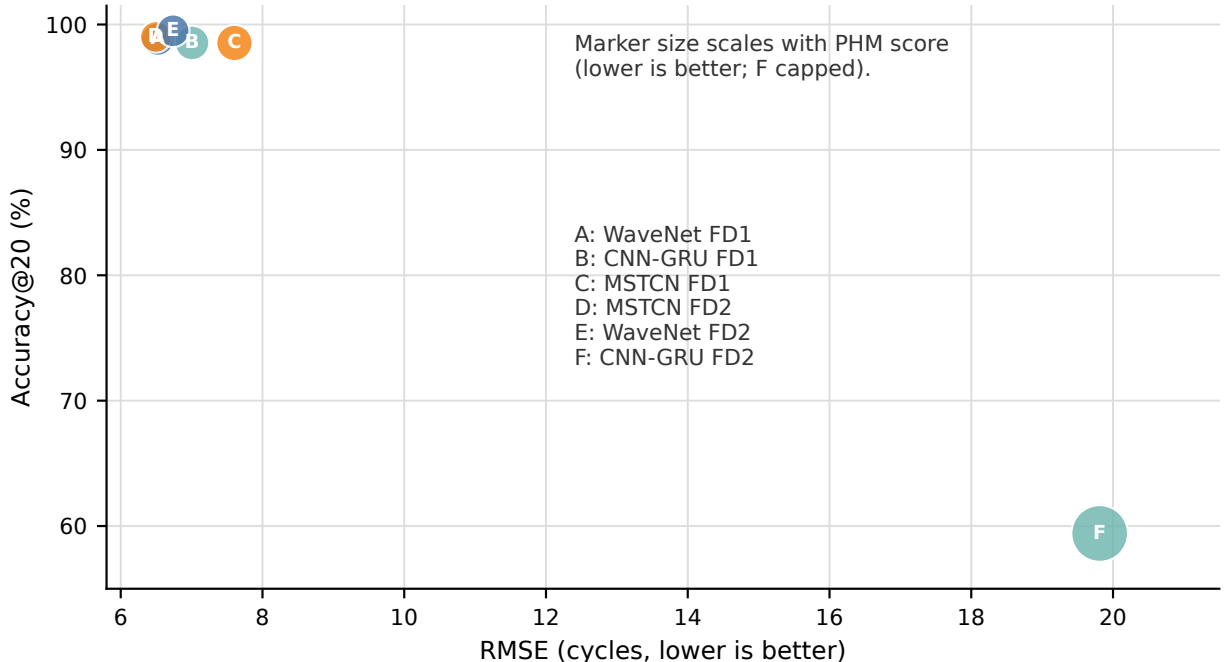


Figure 5: Accuracy–RMSE tradeoff for the controlled FD1/FD2 rows. Marker size scales with normalized PHM score, so lower and smaller points are better on the asymmetric safety metric.

Table 4: Multiseed reproducibility on FD1 ($T=1000$, 30 epochs). Values are mean across available seeds; std is over seeds.

Row	Model	Seeds	Mean RMSE	Std RMSE	Mean R^2
T5	MSTCN	3 (42–44)	8.140	0.74	0.8565
T6	WaveNet	2 (45–46)	8.238	1.44	0.8498

5.2 C2: multi-scale temporal attention

MSTCN is competitive but not uniformly dominant. In the controlled FD1 table, WaveNet ranks first (T1: RMSE 6.523), CNN-GRU second (T2: 7.006), and MSTCN third (T3: 7.604). This contradicts any simple claim that MSTCN is the FD1 winner. On FD2, however, MSTCN is best by RMSE (T11: 6.495), ahead of WaveNet (T12: 6.739), and CNN-GRU degrades sharply (T13: 19.812).

The multiseed runs (Table 4) further bound the single-run estimates: MSTCN’s mean RMSE over three seeds is 8.140 (± 0.74 , T5) and WaveNet’s over two seeds is 8.238 (± 1.44 , T6). At this sample size the means are not meaningfully separable, and both lie above their respective best-run values (T3: 7.604; T1: 6.523), so the FD1 gap between WaveNet and MSTCN in the best-run table should be interpreted alongside seed variability rather than as a stable architecture ordering.

The interpretation supported by the current results is that MSTCN and WaveNet belong to the same top cluster on FD1, MSTCN transfers well to FD2, and full-flight recurrent baselines underperform. The architectural attribution remains open. We have not yet ablated MSTCN’s multi-scale branches, removed its global fusion attention, or matched parameter counts against WaveNet and Transformer. Those experiments are required before claiming that the multi-scale or

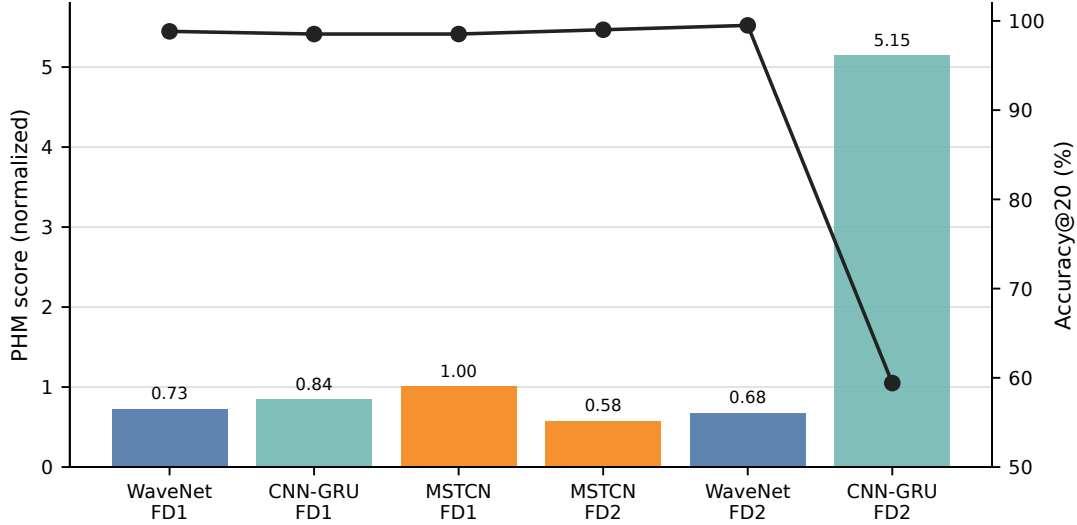


Figure 6: Normalized PHM score and Accuracy@20 for the controlled rows. Most top-cluster rows are tightly grouped on Accuracy@20, while PHM exposes additional separation because it penalizes late predictions asymmetrically.

fusion-attention components themselves cause the observed performance.

5.3 C3: asymmetric loss

The project uses asymmetric MSE as its default loss, and the controlled rows show low PHM scores and high threshold accuracy: WaveNet on FD1 reaches Accuracy@20 98.83 and PHM 0.7256 (T1), and MSTCN on FD2 reaches Accuracy@20 99.01 and PHM 0.5755 (T11). The loss is aligned with the asymmetric PHM scoring tradition [14] and with recent cost-sensitive RUL loss design [11].

However, C3 is not yet an empirical result of this paper. There is no symmetric-MSE baseline in the current canonical results, no α sweep over the late-penalty ratio, and the high-accuracy recipe in the repository uses asymmetric Huber rather than asymmetric MSE (T23: RMSE 9.132, Accuracy@20 93.84). We therefore frame asymmetric MSE as a motivated methodological choice, not as a proven improvement over symmetric MSE in our experiments.

5.4 Open experiments

The next experiments are specific. A single-architecture sequence-length sweep on the top cluster would isolate the C1 effect. MSTCN variants with one dilation scale and with global fusion attention removed would test the C2 mechanism. Symmetric MSE, asymmetric MSE, and asymmetric Huber under identical settings, including an $\alpha \in \{1.0, 1.5, 2.0, 3.0\}$ sweep, would turn C3 from a design choice into an empirical comparison.

6 Conclusion

We presented a deep-learning pipeline for variable-length N-CMAPSS turbofan RUL prediction and used it to compare recurrent, convolutional, hybrid, transformer, and multi-scale attention models under a common preprocessing, training, and evaluation harness. The best controlled FD1 row

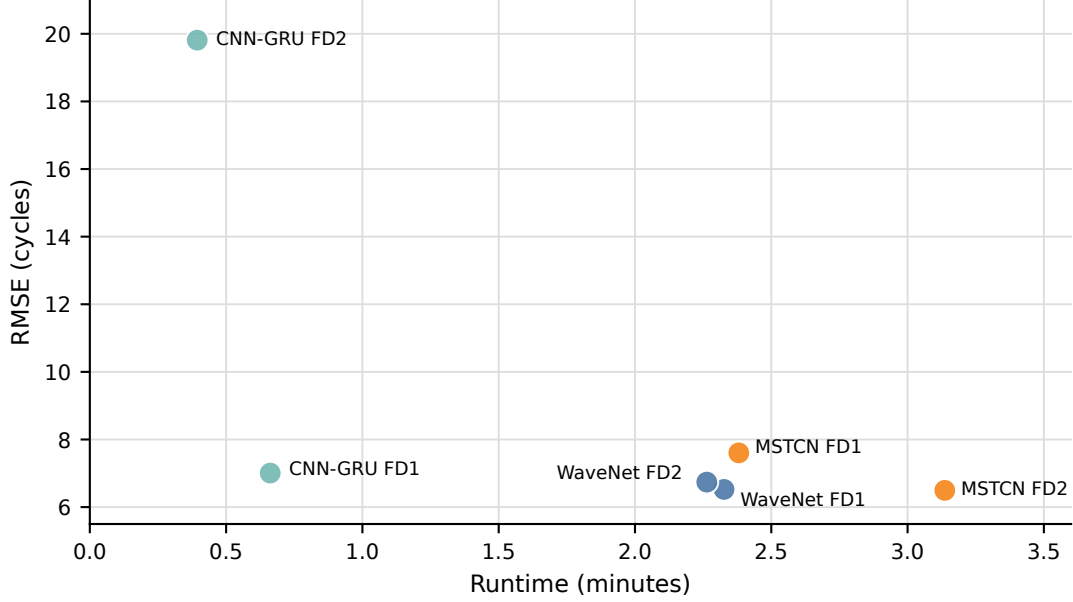


Figure 7: Runtime–RMSE tradeoff for the controlled FD1/FD2 rows. Runtime is measured by the benchmark script on the local machine and is reported as a practical engineering diagnostic, not as a hardware-normalized efficiency result.

reaches RMSE 6.523 (T1), and full-flight recurrent baselines remain far worse (T20: 22.28; T21: 22.20), although this effect is confounded with architecture. MSTCN ranks third on controlled FD1 (T3: 7.604) but first by RMSE on FD2 (T11: 6.495), placing it in the top attention/dilated-convolution cluster rather than above every alternative. Asymmetric MSE is a safety-motivated design choice aligned with the PHM scoring tradition [14] and recent loss-design work [11]; it is not yet a demonstrated empirical improvement over symmetric MSE in this benchmark. The code, configs, benchmark outputs, and W&B traces provide a reproducible basis for extending this comparison.

Limitations

- The sequence-length result is not a clean single-architecture ablation; a matched sweep over 100, 250, 500, 1000, 2000, 5000, and full-flight inputs is needed.
- MSTCN component attribution is incomplete; the multi-scale branches and global fusion attention should be removed independently.
- The asymmetric-loss claim needs a controlled symmetric-MSE baseline, an α sensitivity sweep, and a direct asymmetric-MSE versus asymmetric-Huber comparison.
- Evaluation is currently concentrated on FD1 and FD2; FD3–FD7 remain necessary for broad N-CMAPSS generalization claims.
- Comparisons to STAR, TTSNet, and Xu et al.’s MSTCN are architectural rather than apples-to-apples because those papers report primarily on C-MAPSS, not N-CMAPSS.

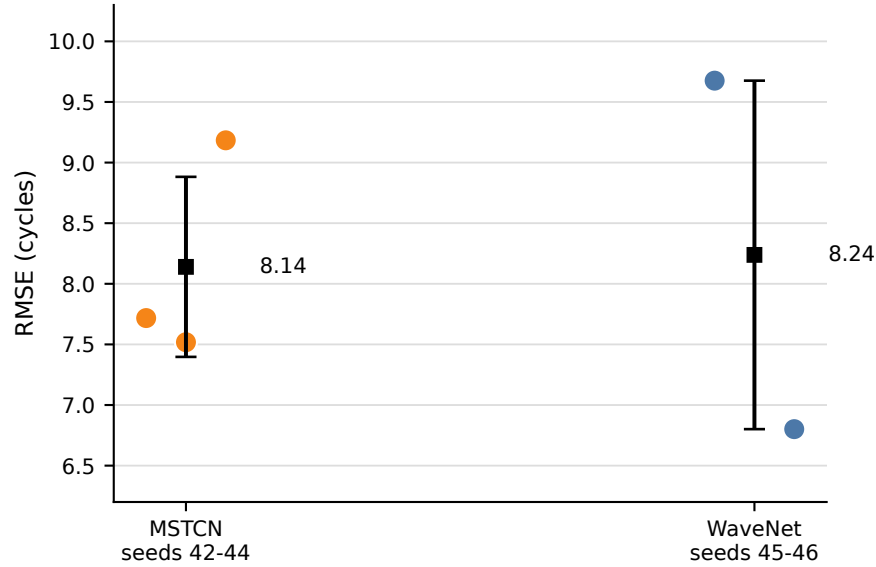


Figure 8: Per-seed RMSE for the FD1 multiseed runs. Dots show individual seeds and black squares show the mean with one standard deviation over the available seeds.

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Table 5: Sequence-length evidence on FD1.

Row	Model	Sequence length	RMSE	R^2	Acc@20
T14	MSTCN	250	10.748	0.7519	91.50
T15	WaveNet	250	11.169	0.7321	96.48
T16	CNN-GRU	250	24.502	-0.2894	46.92
T18	WaveNet	1000	6.523	0.9086	98.83
T17	MSTCN	1000	7.604	0.8758	98.53
T19	TCN	20294	16.13	0.4415	—
T20	LSTM	20294	22.28	-0.0657	—
T21	BiLSTM	20294	22.20	-0.0584	—

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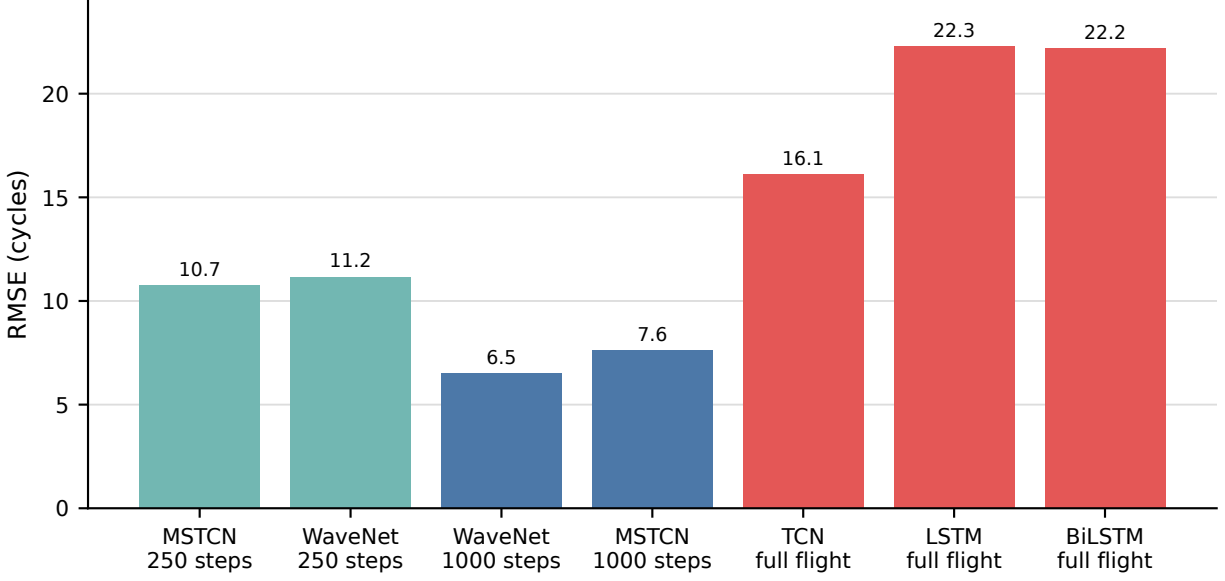


Figure 9: FD1 sequence-length evidence. The 1000-timestep rows are controlled 30-epoch runs, the 250-timestep rows are 10-epoch probes, and the full-flight rows come from older long-sequence baselines; the pattern supports short operational windows but is not a single-factor sequence-length ablation.

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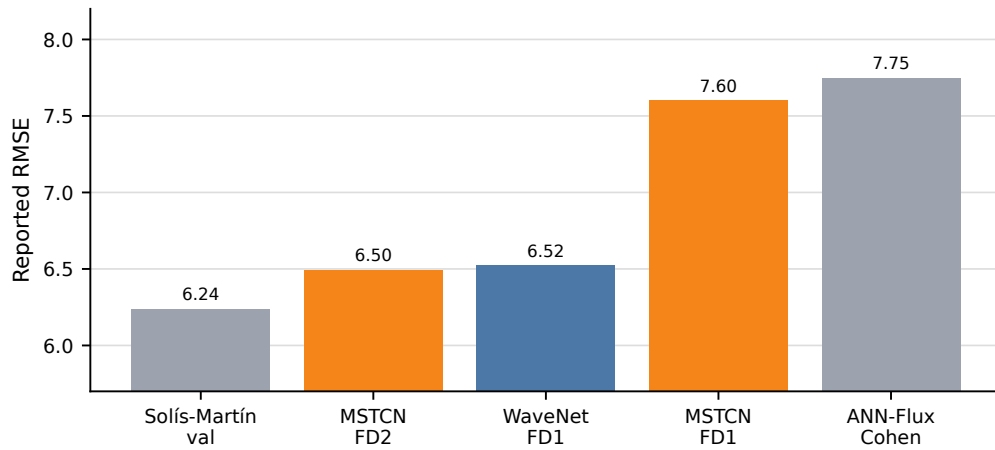


Figure 10: Contextual comparison with the closest N-CMAPSS literature anchors [15, 5]. External rows use their published protocols and are not direct head-to-head baselines; they are included only to show the numerical band of recent N-CMAPSS results.